

🌐 Azure DNS Overview

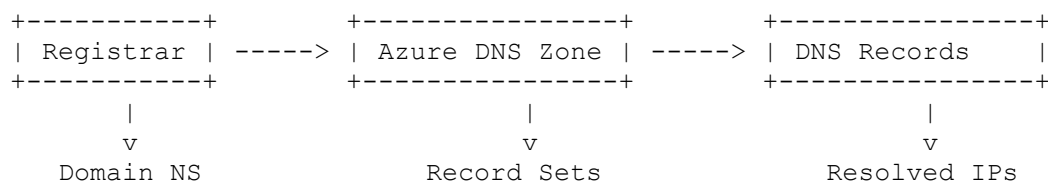
- **Azure DNS** is a hosting service for DNS domains, providing name resolution using Microsoft's global infrastructure.
- It allows you to manage DNS records for your domain without needing your own DNS servers.
- Supports **Public DNS zones** (for internet-facing domains) and **Private DNS zones** (for internal name resolution within VNets).

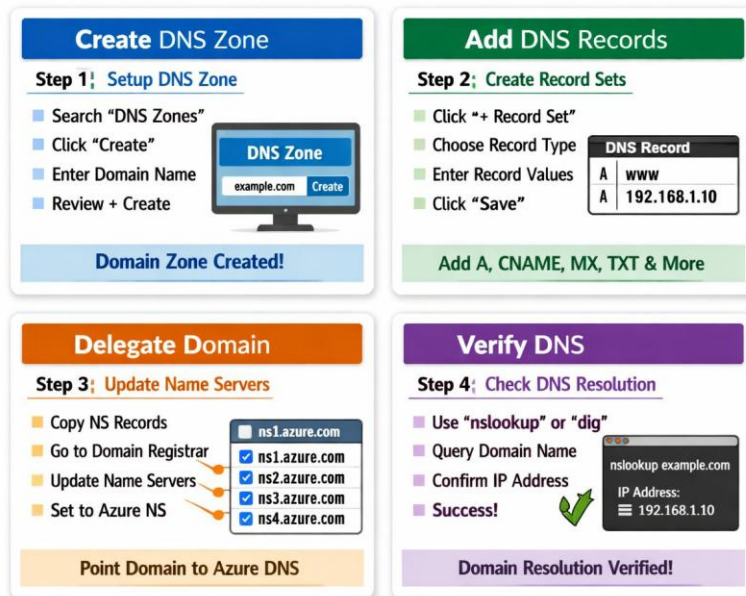
✂ Step-by-Step: Deploy Azure DNS via Azure Portal

1. **Log in to Azure Portal** → portal.azure.com.
2. **Create a DNS Zone:**
 - Search for **DNS zones** → Click **Create**.
 - Select **Subscription** and **Resource Group**.
 - Enter your domain name (e.g., `example.com`).
 - Click **Review + Create** → **Create**.
3. **Add DNS Records:**
 - Open the DNS zone you created.
 - Click **+ Record set**.
 - Choose record type (A, CNAME, MX, TXT, etc.).
 - Enter values (e.g., IP address for A record).
 - Save.
4. **Delegate Domain:**
 - Copy Azure-provided **Name Server (NS) records**.
 - Update them at your domain registrar.
 - This points your domain to Azure DNS.
5. **Verify:**
 - Use `nslookup` or `dig` to confirm resolution.

📊 Diagram: Azure DNS Deployment Flow

Code





⚡ Azure Traffic Manager

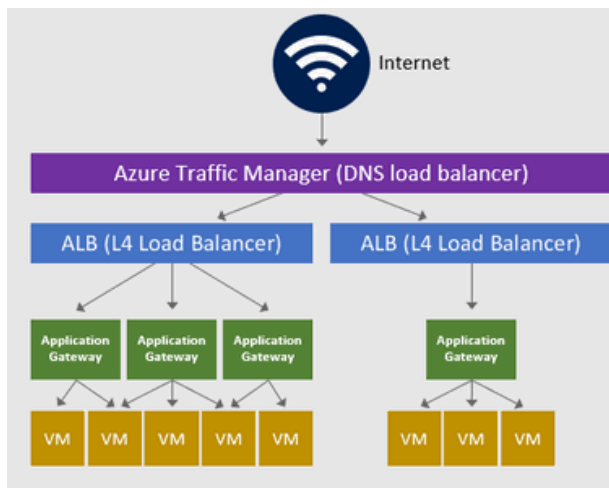
- Traffic Manager** is a **DNS-based load balancer** that distributes traffic across multiple endpoints globally.
- It improves **availability, performance, and failover** by directing users to the best endpoint based on routing policies.

📊 Diagram: Traffic Manager Concept

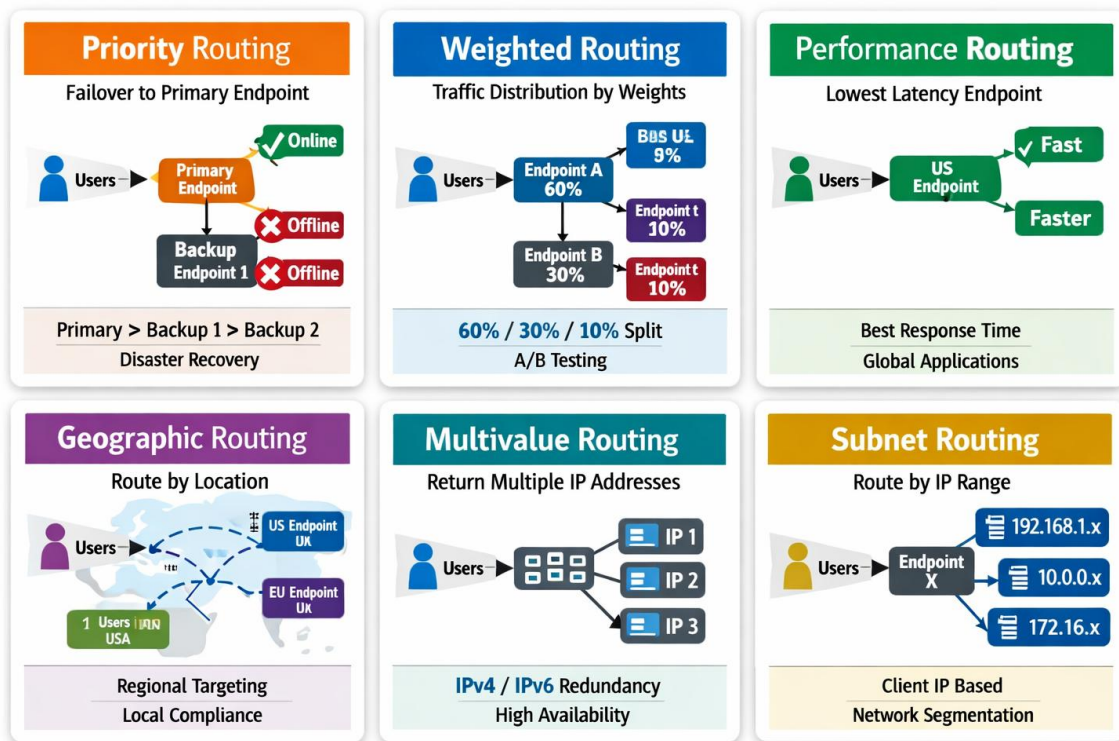
Code

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+-----+ +-----+ +-----+
| Users | ---> | Traffic Manager | ---> | Service Endpoints |
+-----+ +-----+ +-----+
                        | Routing Policy |
  
```



⚡ Azure DNS Routing Policies (Traffic Manager Methods)



Azure Traffic Manager supports **six routing policies**:

Policy	How it Works	Use Case
Priority	Routes to primary endpoint; fails over to backups	Disaster recovery
Weighted	Distributes traffic based on weights	A/B testing, gradual rollout
Performance	Routes to lowest latency endpoint	Global apps needing fast response
Geographic	Routes based on user location	Compliance, localized content
Multivalue	Returns multiple healthy endpoints	Redundancy for IPv4/IPv6
Subnet	Routes based on client IP ranges	Enterprise segmentation

📊 Diagram: Routing Policies Example

```

Code
Users ---> DNS Query ---> Traffic Manager
|         |             |
v         v             v
India --> Geographic --> Endpoint A
US      --> Performance --> Endpoint B
EU      --> Weighted    --> Endpoint C
  
```

✓ Key Takeaways

- **Azure DNS:** Centralized DNS hosting with easy management via Azure Portal.
- **Traffic Manager:** DNS-based load balancer ensuring high availability and performance.
- **Routing Policies:** Six flexible methods (Priority, Weighted, Performance, Geographic, Multivalue, Subnet) to optimize traffic distribution.